

Hsuan-Hao "Howard" Lin

hsuanhal@andrew.cmu.edu ♦ (412)679-2226 ♦ www.linkedin.com/in/hsuanhaolin

EDUCATION

Carnegie Mellon University

Master of Science in Intelligent Information Systems (ML / NLP)

GPA: 4.0/4.0

Pittsburgh, PA

Expected Dec 2026

National Taiwan University

Bachelor of Science in Electrical Engineering

GPA: 4.17/4.3

Dean's List Award: Fall 2020, Fall 2022, Fall 2023

Taipei, Taiwan

Graduated Jun 2024

SKILLS

Programming Languages: Python, C/C++, TypeScript, JavaScript, SQL

Frameworks & Libraries: PyTorch, TensorFlow, LangChain, Scikit-learn, Next.js, tRPC, Sequelize

Tools: Git, Hugging Face, Docker

PROFESSIONAL EXPERIENCE

U-MEDIA Communications Inc.

Part-time Software Engineer

Hsinchu, Taiwan

Sep 2022 – Jul 2025

- Built an in-house payroll system in a 4-member team, automating salary processing for 200+ employees and reducing manual workload, using TypeScript, tRPC, Sequelize and Oracle DB
 - Designed an intuitive dashboard for payroll management and reporting, optimizing user experience based on feedback received from senior management
 - Fulfilled data synchronization between HR and payroll systems, eliminating manual reconciliation and ensuring real-time data consistency
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PROJECTS

Efficient Document Compression in Retrieval-Augmented Generation

Carnegie Mellon University – Course Project

Pittsburgh, PA

Sep 2025 – Dec 2025

- Introduced Semantic Sentence Cover, a lightweight geometric compression method that extracts only the most relevant textual information from retrieved documents to reduce computational and memory costs
- Conducted comprehensive evaluations across multiple QA benchmarks and compression ratios, analyzing the trade-off between context reduction, latency, and answer accuracy
- Achieved 36% end-to-end latency reduction in the RAG pipeline with no significant degradation in QA performance

Dynamic-SUPERB: Instruction-Tuning Benchmark for Speech (ICLR 2025)

National Taiwan University – Research Project

Taipei, Taiwan

Apr 2024 – Jul 2024

- Proposed and applied an event-frequency-weighted sampling strategy to construct a 360-clip DESED subset, maximizing rare event and retained information under limited computational resources
- Implemented F1-score metric, enabling performance comparison for sound event detection tasks
- Integrated DESED task into 180+ tasks benchmark, expanding its evaluation scope for audio event analysis

Audio Captioning Using Text-Only Data

National Taiwan University – Research Project

Taipei, Taiwan

Jan 2024 – May 2024

- Devised a text-only training approach leveraging the CLAP model to align audio and text in a shared representation space, reducing dependence on paired datasets
- Evaluated model performance, achieving a BLEU-1 score of 39 vs. 43.7 with paired data, validating the feasibility of single-modality learning for cross-modal tasks

AV-SUPERB: Audio-Visual Multi-Task Benchmark (ICASSP 2024)

National Taiwan University – Research Project

Taipei, Taiwan

Jan 2023 – Sep 2023

- Developed Audio Event Classification pipeline in 5 tasks benchmark, including preprocessing dataset with over 2.1 million video and model evaluation, contributed key results to benchmark-wide performance report
- Engineered mAP metric with scikit-learn to ensure consistent performance metrics across multi-label classification tasks